ALICE L WALTON SCHOOL OF MEDICINE

Acid Suppression drugs: PPIs and Histamine 2 Receptor Antagonists

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Adapted from: Acid Suppression Therapy

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0:00 / 22:12 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the mechanism of action of H2 Receptor Antagonists (H2RA) and Proton Pump Inhibitors (PPIs).
2. Compare and contrast the pharmacokinetic and pharmacodynamic profiles of H2RA and PPIs taking into consideration their indications, [illegible]
3. Explain the adverse effects of H2RAs and PPIs and discuss possible strategies for minimizing/preventing these effects.
4. Discuss the potential drug interactions associated with H2RA and PPIs and discuss strategies for monitoring and minimizing these risks in clinical practice.

CASE CONNECTION

MB is a 56-year-old female with a history of gastroesophageal reflux disease (GERD) for the past five years. She has been taking omeprazole daily for symptom control. You are seeing her at the clinic on your ECE day, and she expresses concerns about the long-term use of proton pump inhibitors (PPIs) after reading a social media post about potential risks. She has no history of Barrett's esophagus, erosive esophagitis, or other complications of GERD.

You discuss her concerns and suggest transitioning from a PPI to a H2RA, like famotidine. MB is hesitant, asking, "Will switching medications still control my symptoms? And why should I change if the PPI has been working?"

Keep MB in mind as you go through this brick and we will return to her at the end.

GO TO CONCLUSION ↓

How Is Gastric Acid Secreted?

Understanding the process of acid secretion in the stomach is essential to comprehending how [illegible] Proton Pump Inhibitors (PPIs) work. These drugs target different components involved in the acid secretion process to relieve symptoms and promote healing. They are used in the treatment of gastroesophageal reflux disease (GERD), peptic ulcer disease (PUD), gastritis, and others.

Three key substances are involved in stomach acid secretion: acetylcholine (ACh), histamine, and gastrin (Figure 1).

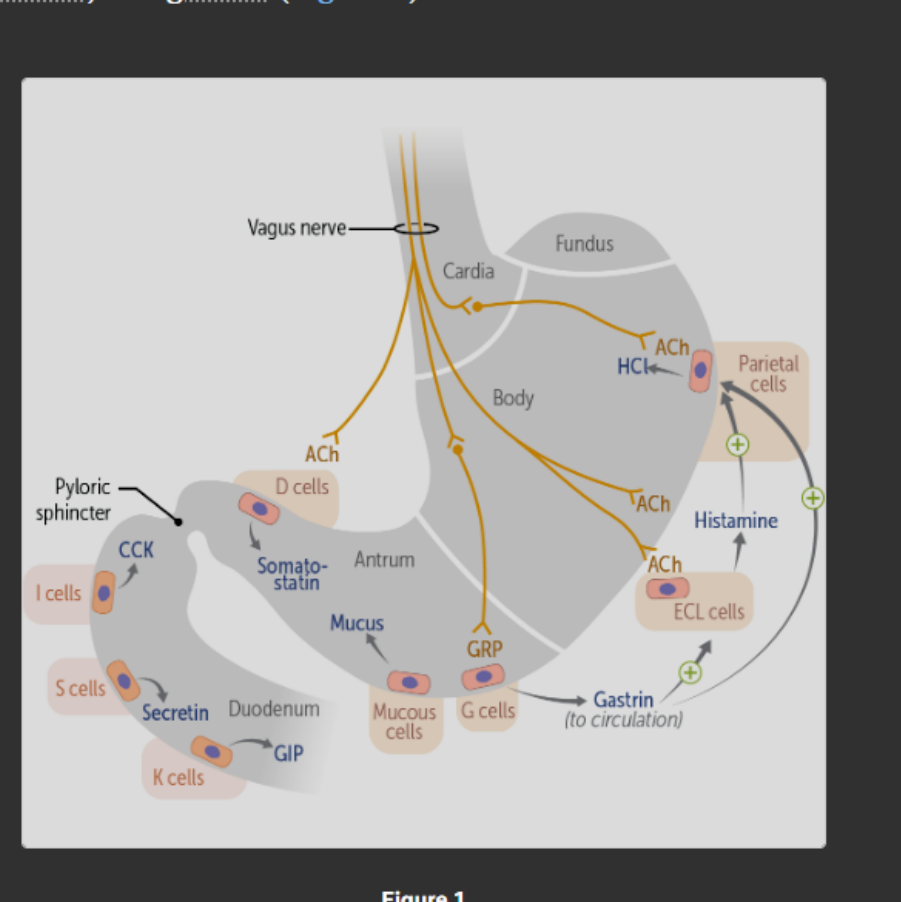


Figure 1

Acetylcholine (ACh) [illegible] binds to M3 receptors on the parietal cells of the stomach. This binding activates the Gq pathway, which leads to an increase of Ca^{2+} and activates H$^+$-K$^+$-ATPase (the proton pump) to release H$^+$.

Postganglionic fibers of the vagus nerve release gastrin-releasing peptide (GRP), which binds to G cells. The G cells then release the hormone **gastrin**, which binds to the gastrin receptor on parietal cells and ECL cells (enterochromaffin-like cells, nearby cells that communicate with parietal cells). Gastrin is also released by distention of the stomach and the presence of partially digested proteins.

Binding of gastrin to the gastrin receptor activates the Gq pathway. In parietal cells, this binding and activation leads to H$^+$ release. In ECL cells, it leads to secretion of **histamine**. The histamine released by ECL cells binds to histamine 2 (H2) receptors on nearby parietal cells. Stimulation of H2 activates the Gs pathway within parietal cells and increases cAMP (cyclic adenosine monophosphate), activating H$^+$-K$^+$-ATPase pumps. Binding of histamine to H2 receptors on parietal cells is the main activator of acid secretion and produces the strongest response.

Somatostatin and prostaglandins inhibit acid secretion by acting on their receptors, resulting in activation of the Gi pathway and a decrease in cAMP.

What are the key substances involved in gastric acid secretion?

How do PPIs work? (LO#1)

Proton Pump Inhibitors

Commonly used PPIs include omeprazole, lansoprazole, esomeprazole, pantoprazole, and rabeprazole (note the -prazole suffix). Omeprazole and esomeprazole are available as prescription (RX) and over the counter (OTC). Pantoprazole is the only PPI that is available intravenously and orally, so it is the most popular choice in the inpatient setting.

PPIs are lipophilic weak bases that reduce the amount of acid produced by parietal cells. They irreversibly **inactivate stomach parietal cell H$^+$-K$^+$-ATPase (proton pump)**, the transporter primarily responsible for producing stomach acid. The irreversible binding produces a longer-lasting effect, and a very potent effect compared to other acid suppression therapy like H2RAs. *Because PPIs inhibit the final pathway for H$^+$ secretion, they inhibit acid secretion in response to all three major stimuli* (ACh, histamine, and gastrin), this is shown in Figure 2.

PPIs are used as first-line medical treatment for GERD, peptic ulcer disease, and gastritis. They are [illegible] e therapy treatment plan used for *H. pylori* [illegible] atment or prophylaxis (e.g., when using high-dose steroids).

What is the mechanism of action of proton pump inhibitors?

How do H2RAs work? (LO#1)

Commonly used H2 receptor antagonists, also called H2 blockers, include cimetidine, famotidine, and ranitidine (note the -dine suffix). Both of these are available OTC, and the only available dosage form is oral.

H2 blockers are a class of drugs that **reversibly block histamine H2 receptors on parietal cells**, inhibiting H$^+$ production. They significantly affect acid secretion stimulated by histamine, especially by inhibiting nocturnal acid secretion, which is mostly mediated by histamine. H2 blockers are effective in treating GERD, peptic ulcer disease, and gastritis and may be used as adjunct in patients with poor re[illegible] PPIs, the H2RAs only inhibit acid production in response to histamine (just as the name suggests). This makes them less potent than the PPIs. Note where H2 blockers exert their action in Figure 2.

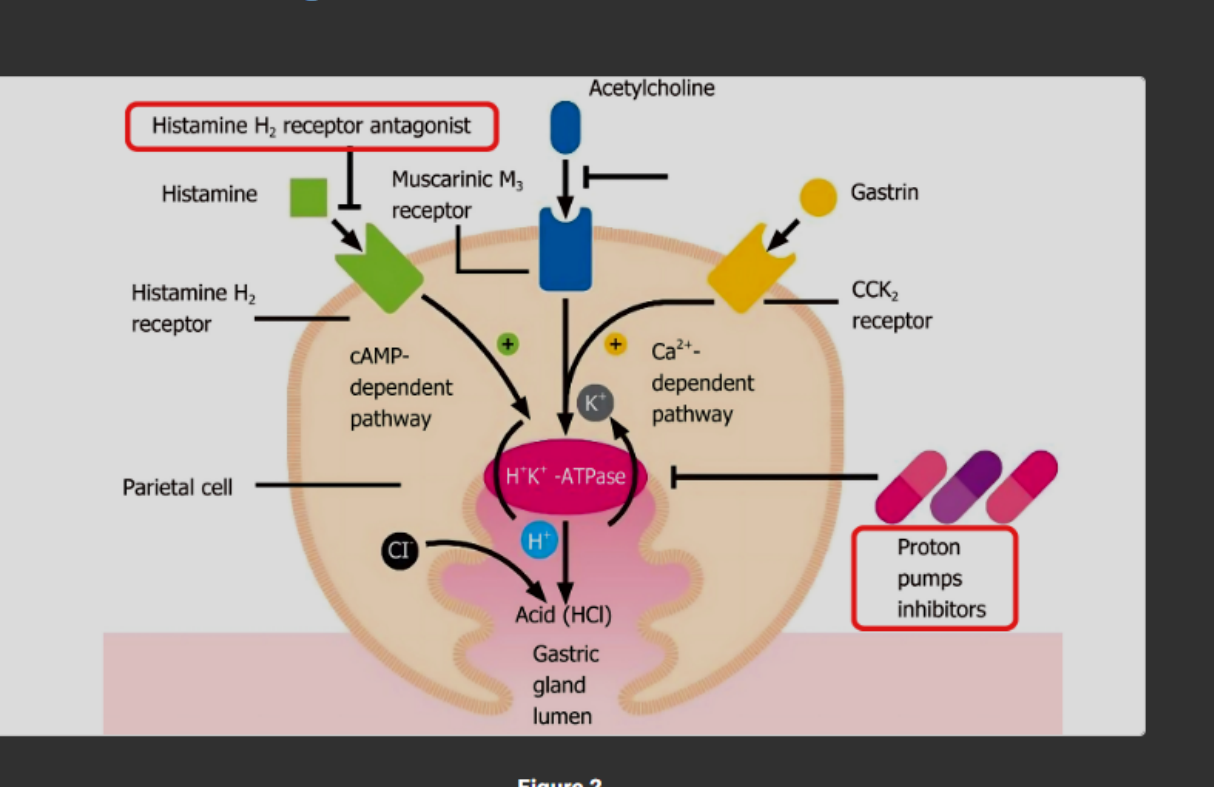


Figure 2

What is the mechanism of action of H2 blockers?

How do H2RAs and PPIs compare? (LO#2)

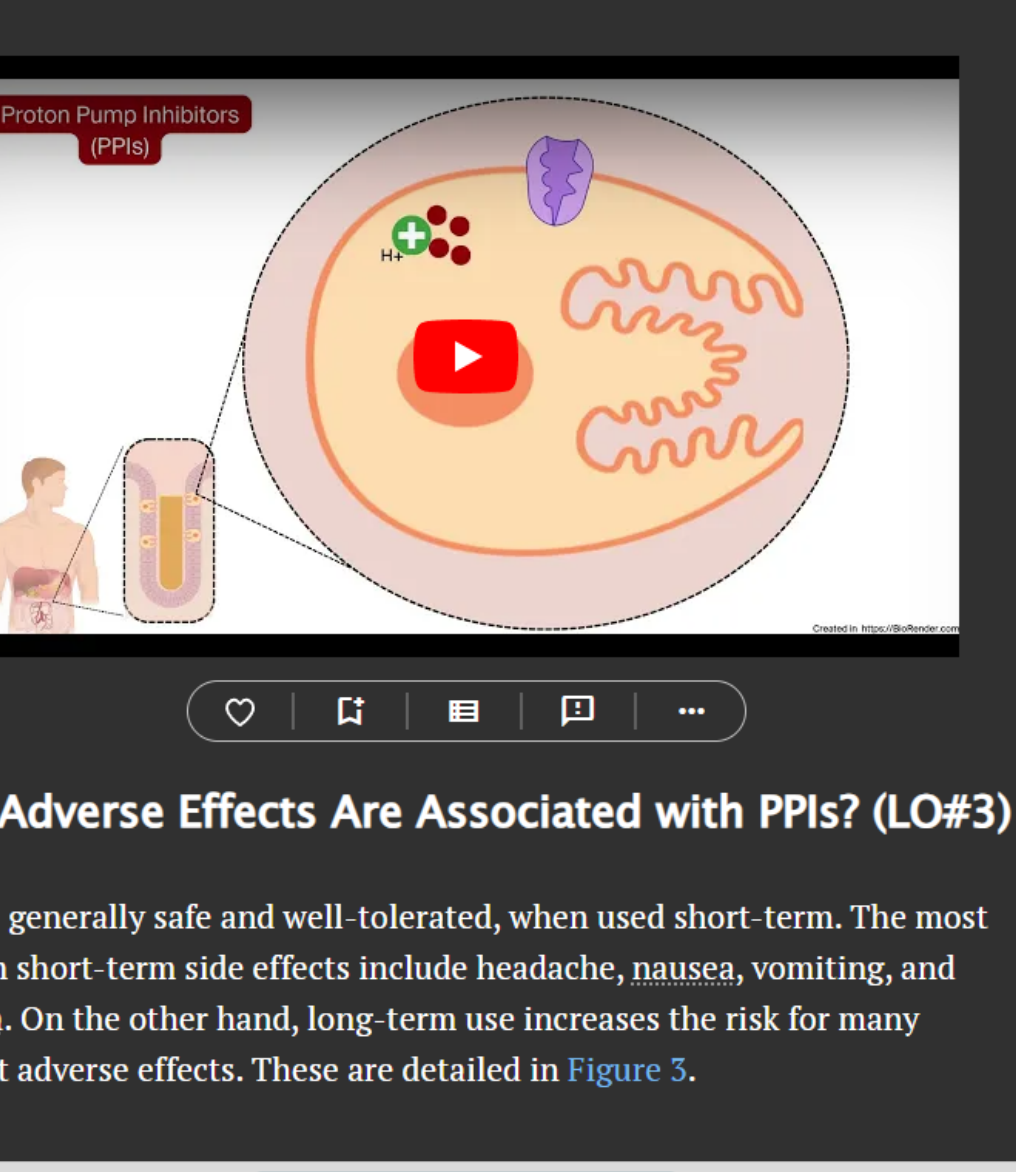
Both H2RAs and PPIs are indicated for heartburn, peptic ulcers, gastroesophageal reflux disease (GERD), and prevention of ulcers in critically-ill patients. For mild cases of heartburn H2RAs are generally preferred, however, when higher amounts of acid suppression are needed PPIs are often preferred. This is related to differences in potency, i.e. how much acid suppression they provide, between both classes of these medications.

CLINICAL CORRELATION

A word about GERD... You will encounter GERD completely in the GI course, but a couple of points are worth mentioning now. Gastroesophageal Reflux Disease (GERD) is a chronic condition where stomach contents flow back (reflux) into the esophagus. This reflux occurs when the *lower esophageal sphincter (LES) weakens or relaxes inappropriately*. As a result, stomach acid irritates the esophageal lining, causing symptoms such as heartburn, regurgitation, and difficulty swallowing. It's not so much of a stomach acid problem, rather than a weakness in the LES. So, by reducing stomach acid production with PPIs, or H2RAs, we can treat GERD.

Recall from above that H2RAs *reversibly block* H2 receptors on parietal cells, which, ultimately, decreases acid production. This more indirect effect stands in contrast to that of PPIs, which *irreversibly inactivate* stomach parietal cell H$^+$-K$^+$-ATPase (proton pump). Because PPIs inhibit the final pathway for H$^+$ secretion, they inhibit acid secretion in response to all three major stimuli (ACh, histamine, and gastrin). In effect, PPIs reduce acid secretion more, i.e., are more potent, and can provide a pharmacodynamic advantage over H2RAs. This means that PPIs are the first choice for some indications, like peptic ulcer disease. However, again thinking of their mechanism, the amount of time PPIs and H2RAs can take to reach their full effect can vary.

When you prescribe a PPI, you educate the patient to take the drug at least 30 minutes before a meal. The video below explains why.



What Adverse Effects Are Associated with PPIs? (LO#3)

PPIs are generally safe and well-tolerated, when used short-term. The most common short-term side effects include headache, nausea, vomiting, and diarrhea. On the other hand, long-term use increases the risk for many different adverse effects. These are detailed in Figure 3.

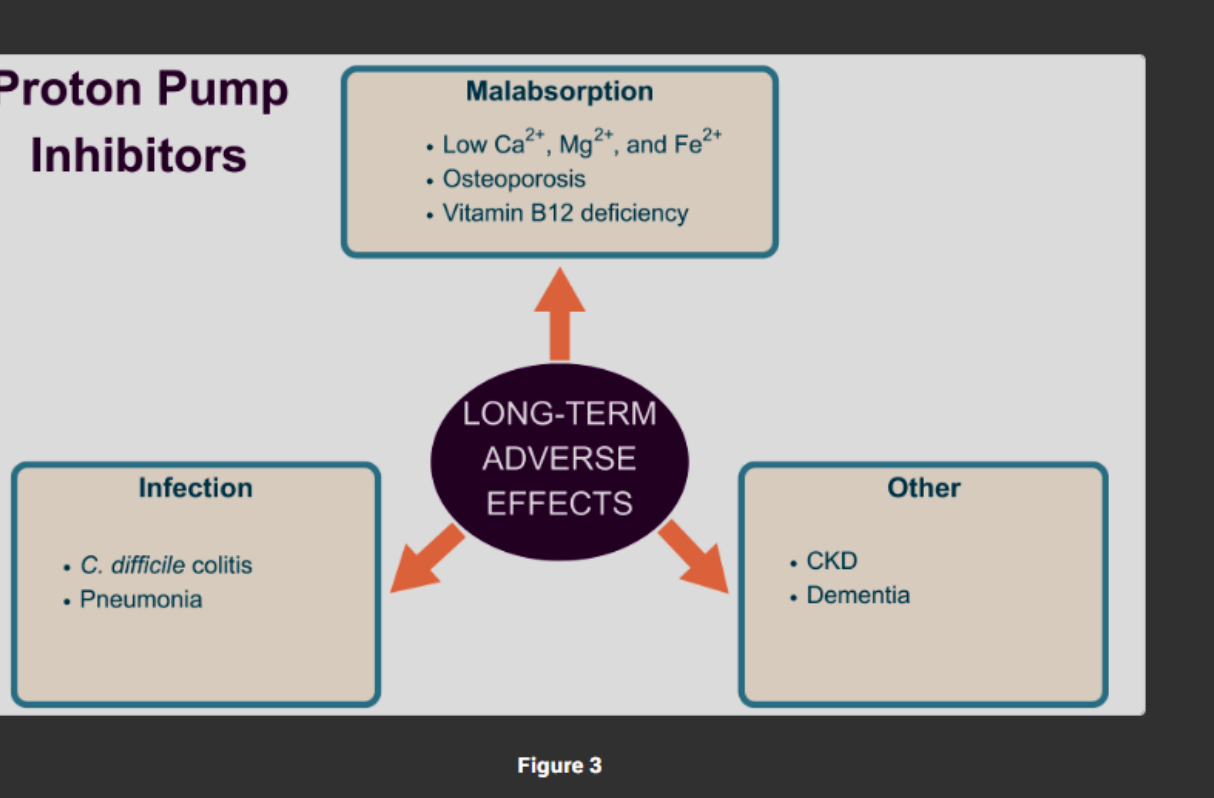


Figure 3

Malabsorption - Given their mechanism on the proton pump, and overall increase in gastric pH (i.e., less acidic), over time PPIs can decrease the absorption of Ca^{2+}, Mg^{2+}, and Fe^{2+} in the gastrointestinal tract. This can lead to an increased risk o[illegible]s in elderly patients. It can also lead to hypomagnesemia, which could be troublesome for patients with a history of cardiac comorbidities and can increase their risk for arrhythmias. PPIs, again because of increase in gastric pH, can alter the absorption of drugs and/ or vitamins. This is particularly noticeable with vitamin B12, as even a low dose of omeprazole may decrease the absorption and result in levels 3-4 times lower than at normal gastric pH.

Infection risk - Patients who take PPIs over an extended period have an increased risk of infection namely, *Clostridium difficile* infection and *pneumonia*. Again, with the altered gastric pH, this allows normal flora to become opportunistic pathogens. It has been found that patients using PPIs for long periods of time have an altered gut microbiota, which, again, lends to increased risk of infection.

Other - Long-term PPI use has also been associated with an increased risk of *chronic kidney disease (CKD)* and *dementia*. The exact mechanism for how this relationship is unclear, however, there is a correlation between long-term PPI use and these two disease states.

Because of the risks associated with long-term use, patients should concurrently initiate lifestyle and behavioral changes (eg, quitting smoking, reducing or stopping alcohol consumption, eating smaller portions, not eating within 3 hours of bedtime) to see if symptoms can be alleviated with time-limited acid suppression therapy. In many cases, patients may be better served with an H2RA, rather than a PPI, and transitioning them may be better for the long-term.

How do PPIs affect the absorption of Ca2+, Mg2+, and Fe2+?

What Adverse Effects Are Associated with H2RAs? (LO#3)

Most H2 blockers are generally safe and adverse effects are rare. The most common side effect of famotidine is headache.

Cimetidine is the exception. Adverse effects include diarrhea, dizziness, headache, and rashes. In addition, it has anti-androgenic effects and can cause prolactin release, resulting in gynecomastia and impotence. Cimetidine crosses the blood-brain barrier and placenta, so it should not be used in during pregnancy. Because of its more extensive adverse effect profile, it is advised to use newer H2 blockers such as famotidine instead of cimetidine.

What are the drug interactions associated with H2RAs and PPIs? (LO#4)

Recall from our previ[illegible] PPIs are metabolized by CYP3A4. This can result in a myriad of drug interactions and altered absorption of the PPIs, or the medications they are interacting with. Caution should always be used when other medications that are substrates of CYP3A34 are used. Additionally, PPIs increase the gastric pH (i.e., less acidic) so they can have effect on the absorption of other drugs. This is particularly pertinent, as mentioned above in the adverse effect section, with vitamin B12 and iron as malabsorption of these substances can lead to anemia.

CLINICAL CORRELATION

There are other medications (e.g., levothyroxine for hypothyroidism) that are greatly affected by the altered gastric pH that PPIs induce, so interactions should always be considered with PPIs.

As mentioned above, the H2 blockers are well tolerated. The exception is cimetidine. It happens to be an inhibitor of CYP3A4, 1A2, 2C19, and 2D6. So, it has interactions with a lot of different medications, and, given that it is an inhibitor of these CYP enzymes, it can cause supratherapeutic, or even toxic, levels of other medications.

Why should caution be used when combining PPIs with other medications?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to MB. Following your expression for her concerns, you would want to discuss the potential long-term effects of PPIs. Although, PPIs provide stronger acid suppression compared to H2RAs, they may carry long-term risks when used indefinitely. This might include malabsorption issues (e.g., osteoporosis) or increased infection risk. H2RAs can be effective for GERD treatment and may be a safer alternative long-term. Switching MB to an H2RA, like famotidine or ranitidine, may be a good option to avoid long-term complications. You would also discuss with MB whether she could benefit from any lifestyle/ behavioral changes, such as quitting smoking, reducing or stopping alcohol consumption, eating smaller portions, and/ or not eating within 3 hours of bedtime.

Summary

- Gastric acid secretion involves acetylcholine, histamine, and gastrin, with histamine being the major stimulus.
- PPIs irreversibly [illegible]ietal cells, reducing acid secretion and are first-line treatment for GERD, peptic ulcer disease, and gastritis.
- H2 blockers reversibly block histamine H2 receptors on parietal cells, inhibiting H+ production and are effective in treating GERD, peptic ulcer disease, and gastritis.
- Long-term PPI use increases the risk of C. difficile infection, pneumonia, fractures, osteoporosis, and hypomagnesemia.
- Cimetidine, an H2 blocker, has a more extensive adverse effect profile and can cause gynecomastia and impotence; newer H2 blockers like famotidine are preferred.
- PPIs and H2 blockers can interact with other medications due to their effects on CYP enzymes and gastric pH, affecting drug absorption and metabolism.

Review Questions

1. Which class of medication provides more potent acid suppression?

A. ○ H2RAs
B. ● PPIs
C. ○ Antacids
D. ○ H2 Blockers
E. ○ Antihistamines

Hide Explanation

The correct answer is PPIs (B). PPIs irreversibly inactivate the proton pump, inhibiting acid secretion in response to all major stimuli, making them more potent. H2RAs (A) reversibly block H2 receptors, providing less potent acid suppression. Antacids (C) neutralize stomach acid but do not suppress acid production. H2 blockers (D) is another term for H2RAs, which are less potent. Antihistamines (E) are not used for acid suppression.

2. Which adverse effect is associated with long-term use of Proton Pump Inhibitors (PPIs)?

A. ○ Reduced risk of C. difficile infection
B. ● Increased risk of osteoporosis
C. ○ Reduced risk of GERD
D. ○ Increased risk of gynecomastia

Hide Explanation

The correct answer is increased risk of osteoporosis (B). Long-term use of PPIs can lead to malabsorption of Ca2+, which can increase the risk of osteoporosis. Increased risk of C. difficile infection (A) is associated with long-term use of PPIs, but it is not the only a[illegible] a benefit of PPIs, not an adverse effect. Inc[illegible]ted with cimetidine, an H2 blocker, not PPIs.

3. What is the mechanism of action of H2 receptor antagonists?

A. ○ They irreversibly inhibit H+-K+-ATPase in stomach parietal cells
B. ● They block histamine H2 receptors on parietal cells
C. ○ They stimulate the release of acetylcholine
D. ○ They inhibit the release of gastrin

Hide Explanation

The correct answer is they block histamine H2 receptors on parietal cells (B). H2 receptor antagonists, also known as H2 blockers, work by reversibly blocking histamine H2 receptors on parietal cells, inhibiting H+ production. They do not irreversibly inhibit H+-K+-ATPase in stomach parietal cells (A), that is the mechanism of action of Proton Pump Inhibitors (PPIs). They do not stimulate the release of acetylcholine (C), that is the function of the vagus nerve in the process of gastric acid secretion. They do not inhibit the release of gastrin (D), gastrin is released by G cells in response to the binding of gastrin-releasing peptide (GRP).

4. Which medication should be avoided during pregnancy?

A. ● Cimetidi[illegible]
B. ○ Famotidine
C. ○ Omeprazole
D. ○ Esomeprazole

Hide Explanation

The correct answer is cimetidine (A). Cimetidine crosses the blood-brain barrier and placenta, making it unsafe for use during pregnancy. Famotidine (B) is a newer H2 blocker with fewer adverse effects. Omeprazole (C) and esomeprazole (D) are PPIs and not specifically contraindicated in pregnancy. Ranitidine (E) is another H2 blocker but not mentioned as unsafe during pregnancy.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describ~~ [illegible] ~~Antagonists (H2RA)~~

blocker but not mentioned as unsafe during pregnancy.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the mechanism of action of H2 Receptor Antagonists (H2RA) and Proton Pump Inhibitors (PPIs).~~

02 ☑ ~~Compare and contrast the pharmacokinetic and pharmacodynamic profiles of H2RA and PPIs taking into consideration their indications, benefits and limitations.~~

03 ☑ ~~Explain the adverse effects of H2RAs and PPIs and discuss possible strategies for minimizing/preventing these effects.~~

04 ☑ ~~Discuss the potential drug interactions associated with H2RA and PPIs and discuss strategies for monitoring and minimizing these risks in clinical practice.~~

Objective Confidence: 4/4